

Assignment 8: Time Series Analysis

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A08_TimeSeries.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Check your working directory
 - Load the tidyverse, lubridate, zoo, and trend packages
 - Set your ggplot theme

```
getwd()
```

```
## [1] "/home/guest/ENV872/EDE_Bella"
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.1      v tibble    3.2.1
## v lubridate  1.9.3      v tidyr     1.3.1
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(lubridate)
library(zoo)
```

```
##
## Attaching package: 'zoo'
##
## The following objects are masked from 'package:base':
##
##      as.Date, as.Date.numeric
```

```
library(trend)
library(ggplot2)
library(here)
```

```
## here() starts at /home/guest/ENV872/EDE_Bella
```

```
#my theme
mytheme <- theme_classic(base_size = 14) +
  theme(axis.text = element_text(color = "black"),
        panel.background = element_rect(fill = "gray95", color = NA),
        plot.background = element_rect(fill = "white", color = NA),
        plot.title = element_text(
          hjust = 0.5,
          face = "bold"),
        legend.position = "right")
```

2. Import the ten datasets from the Ozone_TimeSeries folder in the Raw data folder. These contain ozone concentrations at Garinger High School in North Carolina from 2010-2019 (the EPA air database only allows downloads for one year at a time). Import these either individually or in bulk and then combine them into a single dataframe named **GaringerOzone** of 3589 observation and 20 variables.

```
#1
O3_2010 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_O3_GaringerNC2010_raw.csv"))
O3_2011 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_O3_GaringerNC2011_raw.csv"))
O3_2012 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_O3_GaringerNC2012_raw.csv"))
O3_2013 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_O3_GaringerNC2013_raw.csv"))
O3_2014 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_O3_GaringerNC2014_raw.csv"))
O3_2015 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_O3_GaringerNC2015_raw.csv"))
O3_2016 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_O3_GaringerNC2016_raw.csv"))
O3_2017 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_O3_GaringerNC2017_raw.csv"))
O3_2018 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_O3_GaringerNC2018_raw.csv"))
O3_2019 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_O3_GaringerNC2019_raw.csv"))
```

```
library(dplyr)
```

```
O3_all <- bind_rows(
  O3_2010,
  O3_2011,
  O3_2012,
  O3_2013,
  O3_2014,
```

```
03_2015,  
03_2016,  
03_2017,  
03_2018,  
03_2019)
```

Wrangle

3. Set your date column as a date class.
4. Wrangle your dataset so that it only contains the columns Date, Daily.Max.8.hour.Ozone.Concentration, and DAILY_AQI_VALUE.
5. Notice there are a few days in each year that are missing ozone concentrations. We want to generate a daily dataset, so we will need to fill in any missing days with NA. Create a new data frame that contains a sequence of dates from 2010-01-01 to 2019-12-31 (hint: `as.data.frame(seq())`). Call this new data frame Days. Rename the column name in Days to "Date".
6. Use a `left_join` to combine the data frames. Specify the correct order of data frames within this function so that the final dimensions are 3652 rows and 3 columns. Call your combined data frame GaringerOzone.

```
# 3  
03_all$Date <- as.Date(03_all$Date, format = "%m/%d/%Y")  
  
# 4  
03_all <- 03_all %>%  
  select(Date, Daily.Max.8.hour.Ozone.Concentration, DAILY_AQI_VALUE)  
  
# 5  
Days <- as.data.frame(seq.Date(from=as.Date("2010/01/01"),  
                               to=as.Date("2019/12/31"),  
                               by="day"))  
  
colnames(Days) <- "Date"  
# 6  
GaringerOzone <- left_join(Days, 03_all)
```

```
## Joining with 'by = join_by(Date)'
```

Visualize

7. Create a line plot depicting ozone concentrations over time. In this case, we will plot actual concentrations in ppm, not AQI values. Format your axes accordingly. Add a smoothed line showing any linear trend of your data. Does your plot suggest a trend in ozone concentration over time?

```
#7  
ggplot(GaringerOzone, aes(x=Date, y=Daily.Max.8.hour.Ozone.Concentration))+  
  geom_line()+  
  geom_smooth(method = "lm")+  
  labs(  
    title = "Daily Maximum 8-hour Ozone Concentrations Garinger 2010-2019",
```

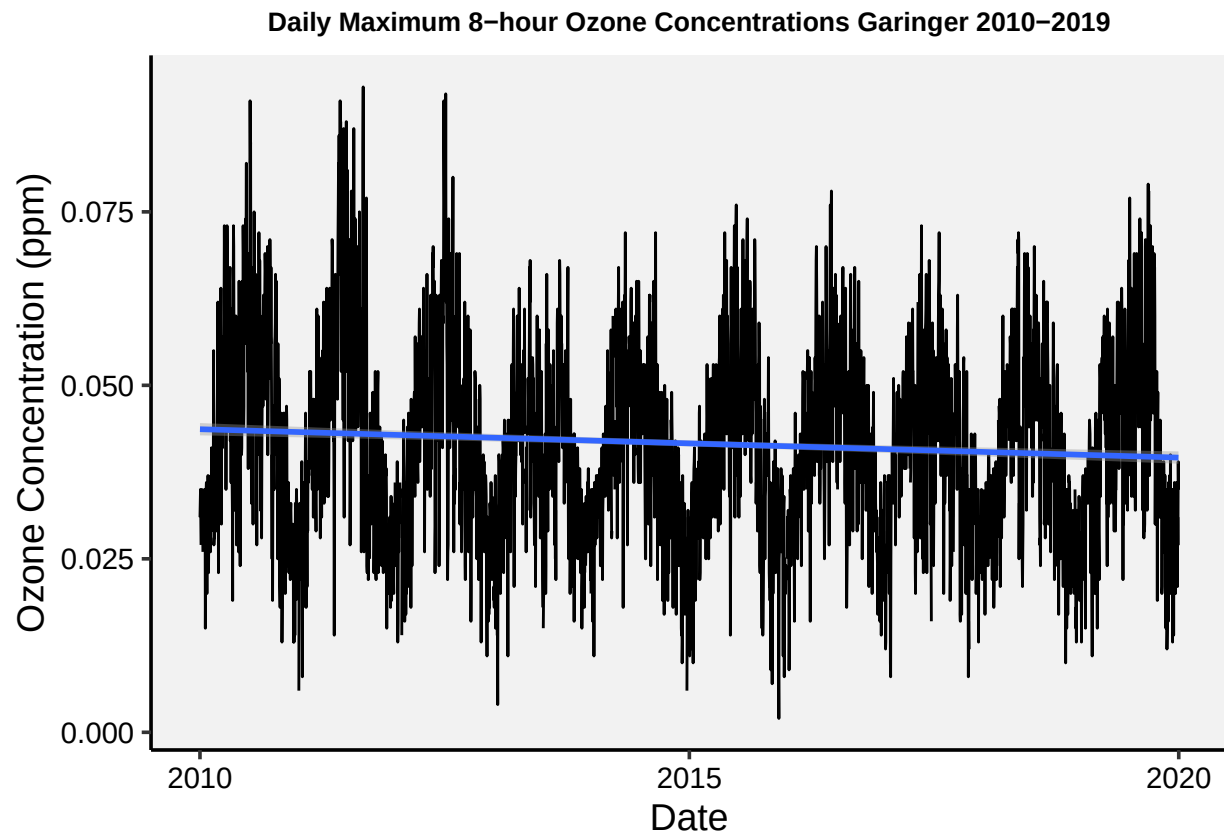
```

x = "Date",
y = "Ozone Concentration (ppm)"
) +
mytheme +
theme(plot.title = element_text(size = 10))

```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 63 rows containing non-finite outside the scale range
## ('stat_smooth()').
```



Answer: My plot shows a slightly descending trend in the daily maximum 8-hour ozone concentrations over 2010 to 2019. It also shows a seasonal variation each year, with higher values in summer and lower values in winter.

Time Series Analysis

Study question: Have ozone concentrations changed over the 2010s at this station?

8. Use a linear interpolation to fill in missing daily data for ozone concentration. Why didn't we use a piecewise constant or spline interpolation?

```
#8
GaringerOzone <-
  GaringerOzone %>%
  mutate(
    O3_interp = zoo::na.approx(
      Daily.Max.8.hour.Ozone.Concentration,
      x = Date,
      na.rm = FALSE
    )
  )
```

Answer: For this dataset, we are studying the concentration of ozone over time, which will change gradually as time goes. Thus, it makes more sense to use linear interpolation, which can give an estimate between existing observations. On the other hand, piecewise constant hold the last available known value of the observation, which is not aligned with the actual data trend. And the spline interpolation would fit a smooth curve through points, which can be unrealistic and over-complicated for our dataset.

9. Create a new data frame called `GaringerOzone.monthly` that contains aggregated data: mean ozone concentrations for each month. In your pipe, you will need to first add columns for year and month to form the groupings. In a separate line of code, create a new Date column with each month-year combination being set as the first day of the month (this is for graphing purposes only)

```
#9
GaringerOzone.monthly <- GaringerOzone %>%
  mutate(Year=year(Date),
         Month=month(Date)) %>%
  group_by(Year, Month) %>%
  summarise(
    mean_ozone = mean(O3_interp)
  ) %>%
  ungroup()
```

'summarise()' has grouped output by 'Year'. You can override using the
'.groups' argument.

```
#create a new Date column
GaringerOzone.monthly <- GaringerOzone.monthly %>%
  mutate(Date = make_date(Year, Month, 1))
```

10. Generate two time series objects. Name the first `GaringerOzone.daily.ts` and base it on the dataframe of daily observations. Name the second `GaringerOzone.monthly.ts` and base it on the monthly average ozone values. Be sure that each specifies the correct start and end dates and the frequency of the time series.

```
#10
#daily
f_day <- yday(first(GaringerOzone$Date))
f_year <- year(first(GaringerOzone$Date))

GaringerOzone.daily.ts <- ts(GaringerOzone$O3_interp,
```

```

start = c(f_year, f_day),
frequency = 365)

#monthly
f_month <- month(first(GaringerOzone.monthly$Date))
f_year_m <- year(first(GaringerOzone.monthly$Date))

GaringerOzone.monthly.ts <- ts(
  GaringerOzone.monthly$mean_ozone,
  start = c(f_year_m, f_month),
  frequency = 12)

```

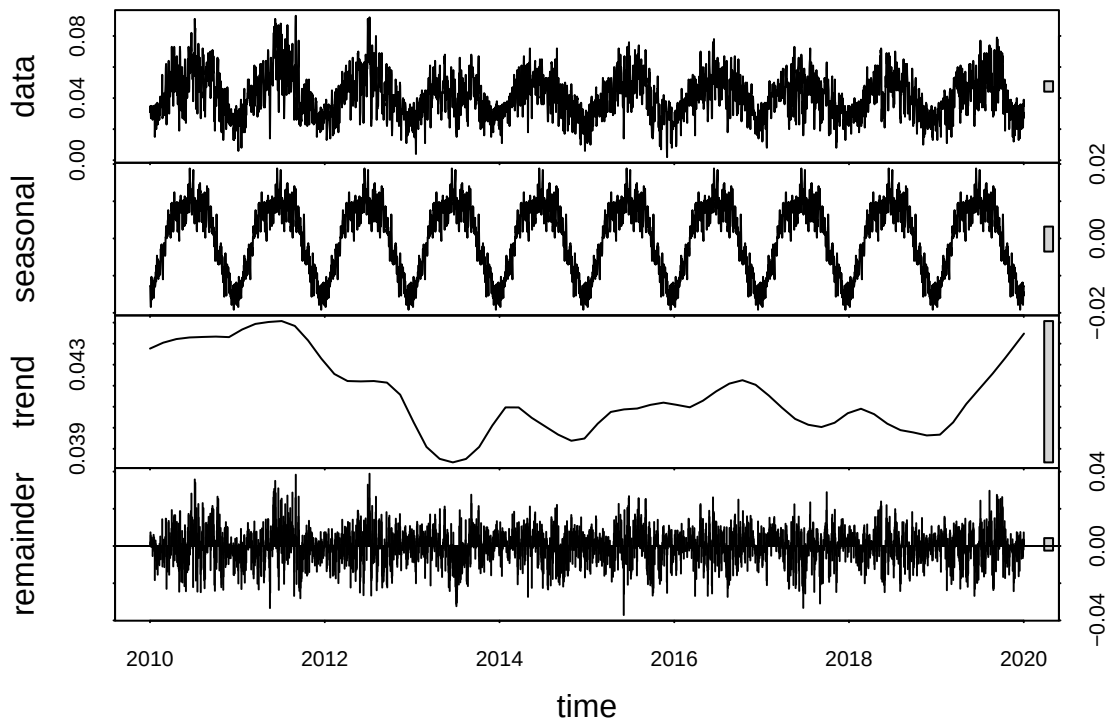
11. Decompose the daily and the monthly time series objects and plot the components using the `plot()` function.

```

#11
O3_daily_Decomp <- stl(GaringerOzone.daily.ts, s.window="periodic")

plot(O3_daily_Decomp)

```

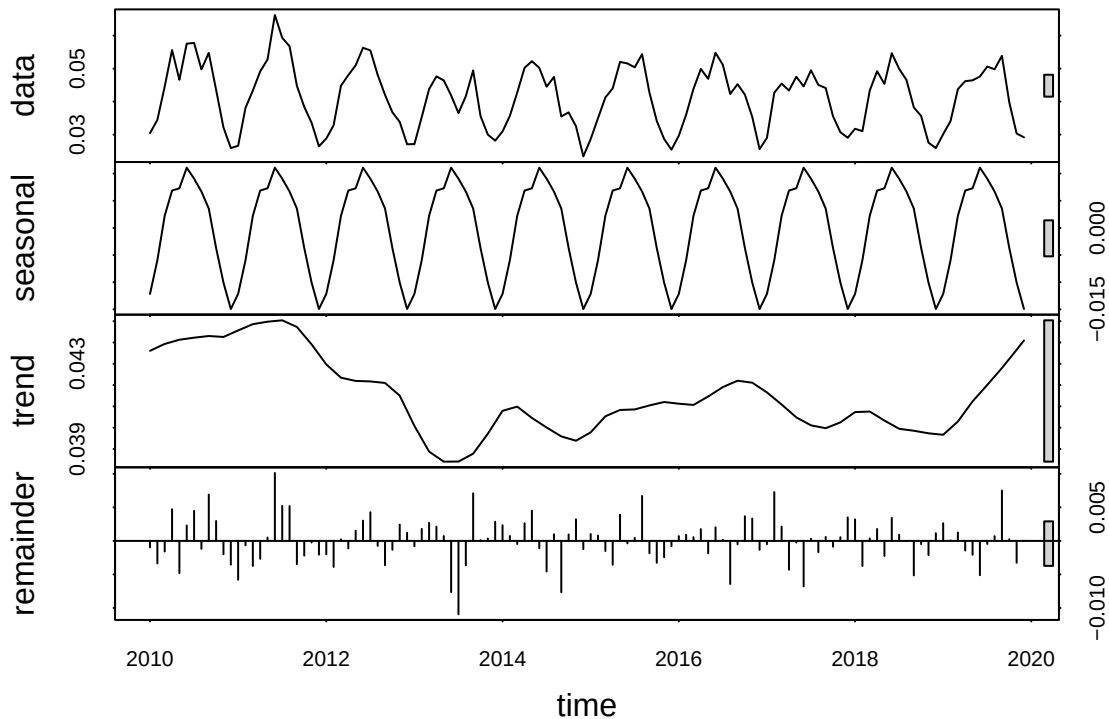


```

O3_monthly_Decomp <- stl(GaringerOzone.monthly.ts, s.window="periodic")

plot(O3_monthly_Decomp)

```



12. Run a monotonic trend analysis for the monthly Ozone series. In this case the seasonal Mann-Kendall is most appropriate; why is this?

```
#12
O3_monthly_trend1 <- Kendall::SeasonalMannKendall(GaringerOzone.monthly.ts)

O3_monthly_trend1

## tau = -0.143, 2-sided pvalue =0.046724

summary(O3_monthly_trend1)

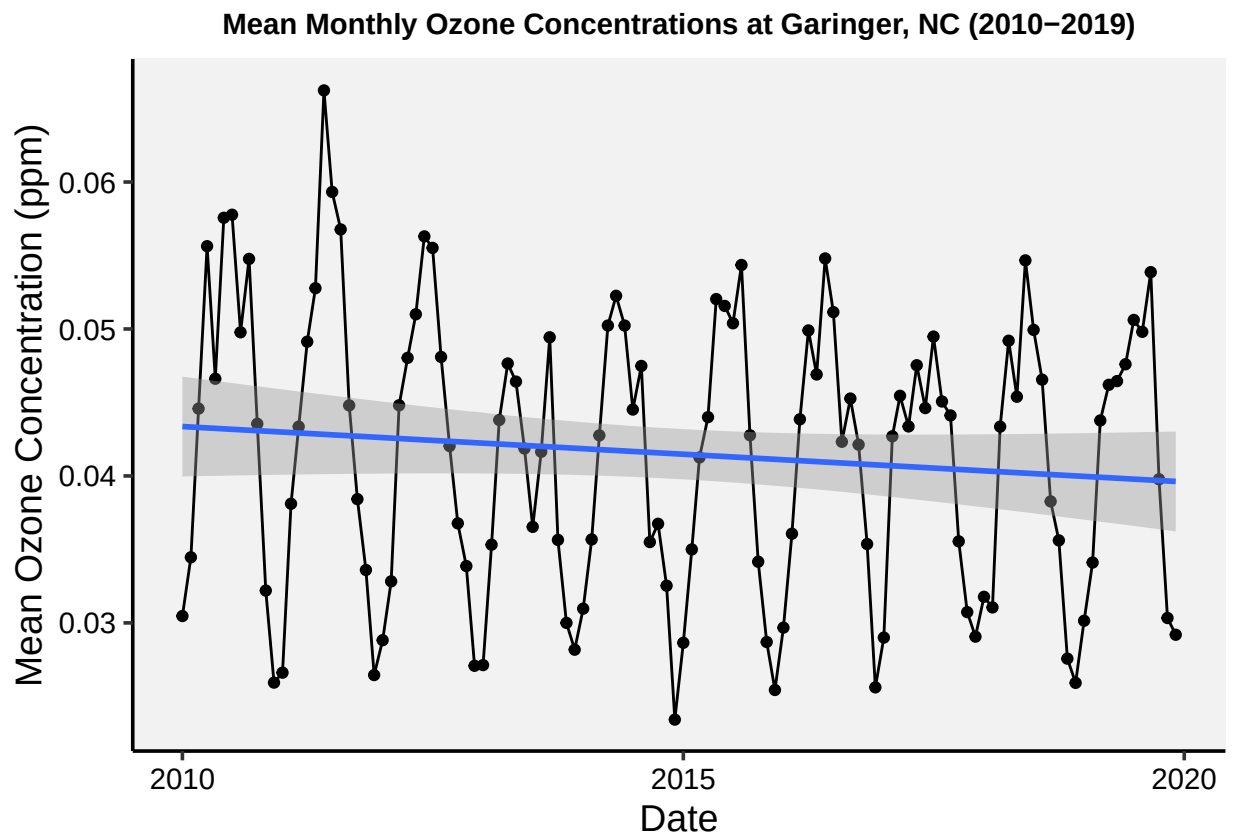
## Score = -77 , Var(Score) = 1499
## denominator = 539.4972
## tau = -0.143, 2-sided pvalue =0.046724
```

Answer: In our case, the seasonal MannKendall is the most appropriate since ozone in our dataset has very strong seasonal cycles. Thus, it is the best option for us to detect the long term monotonic trend with regard of the seasonal bias. Our Tau is -0.143, so there is a weak negative correlation suggesting that there is a slight decline for the ozone level over the 10-year period. The 2-sided pvalue is $0.047 < 0.05$, so this trend is statistically significant at 5% level.

13. Create a plot depicting mean monthly ozone concentrations over time, with both a `geom_point` and a `geom_line` layer. Edit your axis labels accordingly.

```
# 13
ggplot(GaringerOzone.monthly, aes(x = Date, y = mean_ozone)) +
  geom_point() +
  geom_line() +
  labs(
    title = "Mean Monthly Ozone Concentrations at Garinger, NC (2010-2019)",
    x = "Date",
    y = "Mean Ozone Concentration (ppm)"
  ) +
  geom_smooth( method = lm ) +
  mytheme +
  theme(plot.title = element_text(size = 11))
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



14. To accompany your graph, summarize your results in context of the research question. Include output from the statistical test in parentheses at the end of your sentence. Feel free to use multiple sentences in your interpretation.

Answer: The mean monthly ozone concentrations at Garinger shows a strong seasonal trend of fluctuation, with higher values appear in summer and lower values appear in winter. Besides that, over the duration of 2010-2019, there is a weak but statistically significant negative correlation according to our result from the Mann-Kendall test($\tau = -0.143$, 2-sided $p\text{-value} = 0.047$). As a result, the overall ozone level here declines gradually with the consistent seasonal trends.

15. Subtract the seasonal component from the `GaringerOzone.monthly.ts`. Hint: Look at how we extracted the series components for the `EnoDischarge` on the lesson Rmd file.
16. Run the Mann Kendall test on the non-seasonal Ozone monthly series. Compare the results with the ones obtained with the Seasonal Mann Kendall on the complete series.

```
#15
O3_monthly_components <- as.data.frame(O3_monthly_Decomposition$time.series[,1:3])

O3_monthly_components <- O3_monthly_components %>%
  mutate(
    Observed = GaringerOzone.monthly$mean_ozone,
    Date = GaringerOzone.monthly$Date)

O3_monthly_components <- O3_monthly_components %>%
  mutate(
    Deseasonalized = Observed - seasonal)

#16
O3_monthly_deseasonal_ts <- ts(
  O3_monthly_components$Deseasonalized,
  start = start(GaringerOzone.monthly.ts),
  frequency = frequency(GaringerOzone.monthly.ts)
)

O3_monthly_trend2 <- Kendall::MannKendall(O3_monthly_deseasonal_ts)

O3_monthly_trend2

## tau = -0.165, 2-sided pvalue =0.0075402

summary(O3_monthly_trend2)
```

```
## Score = -1179 , Var(Score) = 194365.7
## denominator = 7139.5
## tau = -0.165, 2-sided pvalue =0.0075402
```

Answer: Both the seasonal Mann-Kendall test and the regular Mann-Kendall have a small negative tau value ($\tau_{\text{seasonal}} = -0.143$, $\tau_{\text{deseasonal}} = -0.165$), indicating a weak yet statistically significant declining trend. The pvalue for the seasonal test is 0.0467, which is already showing a significant downward trend with recurring monthly cycles. Now, the pvalue for the deseasonal test is $0.0075 < 0.01$, indicating that the correlation is even more statistically significant. As a result, the ozone concentrations at this site have declined significantly over the decade, and that this trend is not solely driven by seasonal variation but represents an actual long-term improvement in air quality.